FIVE EQUIVALENT CONDITIONS FOR A MORPHISM OF SMOOTH CURVES TO BE SEPARABLE, I.E. ETALE ALMOST EVERYWHERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the differential triangle, the unramified and etale equivalences with automatic flatness, and the function-field criterion with the author's named citation and marked final detail are all retained verbatim.

Editorial note. The label lemma-generically-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Differentials on smooth curves. Morphisms, Lemma morphisms-lemma-smooth-omega-finite-locally-free and morphisms-lemma-triangle-differentials. Makes Omega_{X/k} and Omega_{Y/k} invertible and supplies the triangle f^*Omega_{Y/k} -> Omega_{X/k} -> Omega_{X/Y} -> 0 relating conditions (1) and (2).
- Unramified and etale loci. Morphisms, Lemmas morphisms-lemma-unramified-omega-zero and morphisms-lemma-flat-unramified-etale; Same chapter, Lemma lemma-flat. Equates the support condition with unramifiedness on an open and, flatness being automatic for curves, with etaleness on an open.
- Function fields and generic smoothness. Algebra, Lemma algebra-lemma-smooth-at-generic-point. Identifies the etale-almost-everywhere condition with separability of the function field extension k(X)/k(Y).

Source reference. Original lemma 07ND: sources/stacks-chow/tags.txt, source lines 9758–9758. Prerequisite pointer invoked at proof line(s) 2601; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00HD: sources/stacks-curves/pinned-original-source.tex, source lines 303–303. Prerequisite pointer invoked at proof line(s) 2599; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02GV: sources/stacks-chow/tags.txt, source lines 3242–3242. Prerequisite pointer invoked at proof line(s) 2597; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02G1: sources/stacks-chow/tags.txt, source lines 3212–3212. Prerequisite pointer invoked at proof line(s) 2585; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01UX: sources/stacks-chow/tags.txt, source lines 2426–2426. Prerequisite pointer invoked at proof line(s) 2591; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02G5: sources/stacks-chow/tags.txt, source lines 3216–3216. Prerequisite pointer invoked at proof line(s) 2595; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a morphism of smooth curves over k. The following are equivalent*

(1) *$df : f^*\Omega_{Y/k} \to \Omega_{X/k}$ is nonzero,*
(2) *$\Omega_{X/Y}$ is supported on a proper closed subset of X,*
(3) *there exists a nonempty open $U \subset X$ such that $f|_U : U \to Y$ is unramified,*
(4) *there exists a nonempty open $U \subset X$ such that $f|_U : U \to Y$ is étale,*
(5) *the extension $k(X)/k(Y)$ of function fields is finite separable.*

Proof. Since X and Y are smooth, the sheaves $\Omega_{X/k}$ and $\Omega_{Y/k}$ are invertible modules, see Morphisms, Lemma 02G1. Using the exact sequence

$$f^*\Omega_{Y/k} \longrightarrow \Omega_{X/k} \longrightarrow \Omega_{X/Y} \longrightarrow 0$$

of Morphisms, Lemma 01UX we see that (1) and (2) are equivalent and equivalent to the condition that $f^*\Omega_{Y/k} \to \Omega_{X/k}$ is nonzero in the generic point. The equivalence of (2) and (3) follows from Morphisms, Lemma 02G5. The equivalence between (3) and (4) follows from Morphisms, Lemma 02GV and the fact that flatness is automatic (Lemma 00HD). To see the equivalence of (5) and (4) use Algebra, Lemma 07ND. Some details omitted. □